

CSI Feedback via Joint Source-Channel Coding and Modulation: A Comprehensive Study for 6G Realization

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Abstract— In modern wireless systems, the process of acquiring channel state information (CSI) has become increasingly critical yet challenging, driven by the conflicting needs of expanding antenna arrays and the pursuit of high spectral efficiency. Recent research particularly leveraging joint source-channel coding and modulation (JSCM) via autoencoder architectures, has demonstrated initial promise in significantly enhancing CSI feedback effectiveness. However, a substantial gap persists between algorithm research and practical realization, particularly concerning real-world validation and systematic design methodologies. This article introduces a comprehensive framework enabling JSCM-based CSI feedback, which encompasses a learning methodology robust against wireless channel imperfection and protocol designs tailored for seamless integration into practical cellular systems. A hardware proof-of-concept (PoC) platform is developed and deployed, where experimental results substantiate the proposed method's superiority in realistic wireless environments. Furthermore, by synergizing technical and protocol innovations, this article resolves three critical challenges impeding the deployment of JSCM-based CSI feedback in future 6G networks: variant resource allocation, low peak-to-average power ratio (PAPR) waveforms design, and accurate channel quality indication (CQI) estimation. Equipped with these enabling techniques, the proposed JSCM-based CSI feedback solution is positioned for 6G adoption, demonstrating substantial throughput improvements over conventional separate source and channel coding (SSCC) implementations.

Index Terms—Channel state information (CSI) feedback, 6G, joint source-channel coding and modulation (JSCM), uplink control channel, autoencoder, prototype, proof-of-concept (PoC)

I. INTRODUCTION

High-speed downlink services in modern wireless systems rely on precoding to eliminate signal distortion from the channel. This procedure inherently requires accurate downlink channel state information (CSI), which is acquired through user equipment (UE) feedback via uplink control signaling. However, the continuous demand for higher data rates prompts mobile systems to evolve towards a paradigm with more antennas [2], which aggravates uplink overhead for the downlink CSI sharing, leading to the loss of spectral efficiency. This

issue scales linearly and can lead to a complete occupation by overhead in the uplink channel when the antennas increase to 256 (i.e., a plan of upper mid-band scenario) [3]. Therefore, developing efficient CSI feedback mechanisms appears to be imperative in unlocking the complete potential of mobile systems.

Existing solutions for CSI feedback predominantly employed codebook-based compression, exemplified by type-I, type-II, and enhanced Type-II (e-type-II) in 5G new radio (NR), primarily to reduce Uplink Control Information (UCI) overhead [4]. However, such compression methods inherently sacrifice CSI accuracy due to limited granularity, consequently leading to diminished downlink throughput. To address this limitation, recent studies [5–10] and standardization efforts in 3rd generation partnership project (3GPP) [11, 12] have investigated leveraging artificial intelligence (AI) for CSI feedback. It is essentially a source compression and reconstruction problem, making the autoencoder architecture an ideal framework for this task [13]. Inherently, autoencoders can capture the intrinsic correlations of CSI in an end-to-end manner, thereby yielding superior compression performance [10]. This approach was initially applied to massive MIMO CSI feedback in [5], with subsequent studies [6–8] further refining the neural network architectures and algorithmic frameworks. Besides, transitioning from conventional codebook-based solutions to autoencoder-driven architectures introduces more compatibility challenges, such as quantization mismatch and channel quality indication (CQI) estimation difficulty [11]. Recent literature has begun to address these issues, for instance, [9] employs a proxy neural network at the UE side to predict the reconstructed CSI for more precise CQI estimation.

Nevertheless, this aforementioned solution solely focuses on source compression within the conventional separate source and channel coding (SSCC) framework. This separation limits its information-carrying capability and suffers from “cliff-effect” under varying channel conditions, ultimately degrading performance. A promising alternative to overcome these drawbacks is the deep joint source and channel coding (D-JSCC) paradigm. By integrating source and channel coding into a single optimizable neural network, D-JSCC has demonstrated considerable potential in semantic communication tasks involving diverse data modalities, such as text [14–16], audio [17, 18], as well as image and video transmission [19–23]. Concurrently, these pioneering works have validated the remarkable efficacy of autoencoder architectures in D-JSCC tasks, consistently outperforming SSCC baselines, with diverse backbone models, such as convolutional neural networks (CNNs) [17, 19, 20] and Transformers [15, 18]. The works most closely related to this paper are [21, 22], which

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propose JSCC approaches compatible with digital modulation. Moving beyond coding, the autoencoders in [21, 22] are tasked with optimizing the probability distribution of the modulated symbols, realizing probabilistic constellation shaping.

Inspired by the potential demonstrated by semantic communication approaches, utilizing D-JSCC strategy to facilitate CSI feedback has been proposed by [24]. This work showcased significant performance benefits, notably in enhancing CSI reconstruction accuracy and alleviating the cliff effect common in conventional methods. The framework introduced in [24] features an encoder that directly maps the input CSI to complex-valued modulated symbols. These modulated symbols' real and imaginary parts are then directly used as the in-phase and quadrature components for transmission. This represents a paradigm shift from conventional D-JSCC approaches, which typically output coded bits subsequently fed into a separate, standard modulation mapping block. In contrast, the encoder directly shapes the modulated symbols, resulting in a continuous distribution within the constellation. This distribution approximates Gaussian characteristics for power and uniform characteristics for phase, which is known from information theory to approach the Shannon channel capacity. Consequently, this method yields considerable performance gains compared to other AI-based CSI feedback techniques relying on separate coding and modulation. To clearly distinguish this integrated design, we refer to it as a joint source-channel coding and modulation (JSCM) scheme, and the prefix 'D' is omitted, acknowledging that non-neural network implementations for this specific integrated task are not prevalent in prior work.

Building upon this foundation, subsequent works such as [25–27] have further advanced the JSCM framework. These studies explored variations including single-ended neural network implementations, joint CSI feedback and precoding design, and the integration of pilot design, channel estimation, CSI feedback, and precoding within a unified model. Collectively, this body of research underscores the significant capability of the JSCM framework to enhance CSI feedback performance compared to conventional SSCC methods, suggesting the potential for end-to-end neural networks to replace multiple traditional processing blocks. Despite these promising results, significant hurdles remain in integrating JSCM into standard cellular protocols and ensuring its coexistence with established data communication pathways. This is primarily attributable to the bit-centric nature of existing physical layer protocols. Moreover, much of the current JSCM literature concentrates on link-level optimization, often neglecting crucial system-level considerations and practical implementation challenges. For instance, from a link perspective, JSCM models often trained assuming additive white Gaussian noise (AWGN) channels must prove robust to real-world impairments like residual fading effects and radio frequency (RF) imperfections. Their practical feasibility thus requires validation through field trials, not just simulations. From a system perspective, viable JSCM solutions must feature models adaptable to dynamic resource allocation, generate waveforms meeting stringent peak-to-average power ratio (PAPR) requirements, and include a mechanism of CQI estimation to inform the channel quality

of the downlink channel impact by the reconstructed CSI to the Base Station (BS).

This article aims to bridge the gap between the theoretical research on JSCM for CSI feedback and its practical realization within cellular systems, thereby paving the way for its potential integration into future 6G networks. We present a comprehensive investigation into JSCM-based CSI feedback, addressing key aspects including technical design considerations, necessary protocol adaptations, thorough performance evaluations, and practical field validation. The principal contributions of this work are summarized as follows:

- We propose a holistic framework for JSCM-based CSI feedback that integrates two key innovations, including: 1) A robust learning strategy that compensates for residual fading and RF impairments during training, enhancing resilience against real-world channel distortions; and 2) A redefined physical uplink control channel (PUCCH) structure that inherently, ensuring compatibility with cellular protocol stack requirements while maintaining system configuration flexibility.
- A hardware proof-of-concept (PoC) platform compliant 5G NR configurations is developed and tested in a crowded indoor office scenario, demonstrating reliable deployment of JSCM-based CSI feedback under real-world propagation conditions. This experimental validation confirms the method's operational viability in current infrastructure.
- One critical challenge lies in handling resource allocation variability in real systems that requires JSCM models to produce outputs with matching lengths. To this end, we propose a method integrating a latent-hierarchical model with priority-aware rate matching in PUCCH. Specifically, the encoder organizes latent features that encapsulate CSI content with different priority. During transmission, PUCCH maintain retention of model output according to priority that enables a single model serving arbitrary resource.
- The second critical challenge that obstructs JSCM-based CSI feedback adoption are the Gaussian-like constellation distribution that inherently elevates PAPR. To resolve this, we develop a modulation shaping method that strategically reallocates information between phase and amplitude. This trade-off balances PAPR performance and sacrificed feedback accuracy, that provides a re-configurable PAPR profile to meet any power amplifier requirements.
- The final critical challenge that is the UE cannot readily compute the effective downlink channel gain due to lacking the knowledge of the potential CSI reconstruction inaccuracy. To overcome this impediment, we introduce a CQI estimation methodology, where the UE emulates decoding procedure of the BS with a latent representation corrupted by artificially synthesized noise. This process yields an estimate of the CSI as it would likely be reconstructed at the BS enabling the CQI calculation.

The rest of the paper is organized as follows. Section II provides the problem formulation and system model. Section III proposes framework, learning methodology, and protocol

impacts of JSCM-based CSI feedback. Section IV presents the developed PoC and experimental results. Section V proposes three critical enablers of JSCM-based CSI feedback, including variant-resource adaptation, low PAPR waveform, and CQI estimation. Finally, Section VI concludes the work.

II. PROBLEM FORMULATION & SYSTEM MODEL

We consider a massive MIMO system with $N_{bs} > 1$ antennas at a BS, and $N_{ue} \geq 1$ antennas at a UE. The BS schedules a bandwidth for downlink transmission experiencing a channel represented by $\mathbf{H}_d \in \mathbb{C}^{N_{ue} \times N_{bs}}$. The received signal vector at the UE, $\mathbf{y}_d \in \mathbb{C}^{N_{ue} \times 1}$, is expressed as

$$\mathbf{y}_d = \mathbf{H}_d \mathbf{W} \mathbf{x}_d + \mathbf{n}_d, \quad (1)$$

where $\mathbf{W} \in \mathbb{C}^{N_{bs} \times N_{ue}}$ represents the precoding matrix, $\mathbf{x}_d = [x_1, x_2, \dots, x_{N_{ue}}]^T$ is a vector including modulated symbols carrying downlink data, and $\mathbf{n}_d \sim \mathcal{CN}(0, \sigma^2 \mathbf{I}_{N_{ue}})$ denotes the AWGN with zero mean and variance σ^2 . Considering that channel reciprocity is invalid, a CSI feedback procedure is required for obtaining the precoding matrix $\mathbf{W}$ at the BS. Briefly speaking, the BS allocate and transmits CSI reference signal (CSI-RS) in the downlink bandwidth to measure downlink CSI, and then the UE transmits the measured downlink CSI back to BS via PUCCH. While the proposed methodology and evaluations are equally applicable to the physical uplink shared channel (PUSCH), the subsequent discussion focuses exclusively on the PUCCH for the sake of brevity.

We consider the content of downlink CSI is represented as $\Omega = [\omega_1, \omega_2, \dots, \omega_{N_f}]$, where $\omega_i \in \mathbb{C}^{N_{ue} \times N_{bs}}$ corresponds to the CSI of the i th frequency unit, and N_f denotes the number of frequency units configured as sub-band or full-band granularity depending on BS scheduling. The format of each CSI unit ω_i can be either a precoding matrix $\mathbf{W}_i$ or a measured channel matrix $\hat{\mathbf{H}}_{d,i} \in \mathbb{C}^{N_{ue} \times N_{bs}}$ for the i th frequency unit. Accordingly, upon receiving the feedback, the BS either directly utilize the received $\mathbf{W}_i$ or compute the precoding matrix based on the received channel estimate $\hat{\mathbf{H}}_{d,i}$. In this work, we do not restrict the specific format of CSI.

After measurement, the CSI is sent back to the BS via the allocated resource on PUCCH. To do so, a symbol vector $\mathbf{s} \in \mathbb{C}^{k \times 1}$ that carries downlink CSI Ω can be expressed as:

$$\mathbf{s} = f_{\text{encode}}(\Omega, \mu), \quad (2)$$

where $\mathbf{s} = [s_1, s_2, \dots, s_k]^T$ is the vector including k number of modulated symbols carrying CSI information, f_{encode} represents an encoding functions for obtaining the symbol vector $\mathbf{s}$, and μ is the estimated SNR. Then, a PUCCH procedure is executed, including the necessary procedures for transmitting $\mathbf{s}$, such as rate-matching, interleaving, and resource mapping. Accordingly, a symbol vector $\mathbf{s}_{\text{PUCCH}} \in \mathbb{C}^{k \times 1}$ that is mapped to several sub-carriers to be transmitted, which can be expressed as:

$$\mathbf{s}_{\text{PUCCH}} = f_{\text{PUCCH}}(\mathbf{s}, \Psi), \quad (3)$$

where f_{PUCCH} represents the PUCCH function that maps modulated symbols to exact orthogonal frequency-division multiplexing (OFDM) resource element (RE), and Ψ is the allocated channel resource.

Considering the uplink channel experienced by the PUCCH for CSI feedback is represented by $\mathbf{H}_u \in \mathbb{C}^{N_{bs} \times 1}$. Given that PUCCH uses single-layer transmission, the channel model for uplink is assumed with only one UE-antenna, despite the presence of multiple physical antennas. The received feedback at the BS is expressed as

$$\hat{\mathbf{s}}_{\text{PUCCH},i} = \phi^H(\mathbf{H}_u \mathbf{s}_{\text{PUCCH},i} + \mathbf{n}), \quad (4)$$

where ϕ^H represents the conjugate transpose of ϕ , $\phi = \mathbf{H}_u / \|\mathbf{H}_u\|_2$ is the combining vector, and $\mathbf{n}$ is the noise. We consider that the procedures of baseband and RF signal processing are omitted. The vector $\hat{\mathbf{s}}_{\text{PUCCH}} = [\hat{s}_{\text{PUCCH},1}, \hat{s}_{\text{PUCCH},2}, \dots, \hat{s}_{\text{PUCCH},k}]^T$ denotes the complex values for each subcarrier recovered after channel equalization at the BS. Then, the modulated symbols carries downlink CSI $\hat{\mathbf{s}}$ can be gathered by

$$\hat{\mathbf{s}} = f_{\text{PUCCH}-r}(\hat{\mathbf{s}}_{\text{PUCCH}}). \quad (5)$$

Here, $f_{\text{PUCCH}-r}$ denotes the reverse PUCCH procedure at the BS, which involves de-mapping, de-interleaving, and de-multiplexing the complex values from sub-carriers. Finally, the CSI can be reconstructed via decoding function f_{decode} according to

$$\hat{\Omega} = f_{\text{decode}}(\hat{\mathbf{s}}, \mu). \quad (6)$$

This work targets maximizing the accuracy of reconstructed CSI (i.e., equivalent to downlink throughput) under limited uplink overhead by designing the algorithms f_{encode} and f_{decode} , as well as the paired PUCCH procedure f_{PUCCH} and $f_{\text{PUCCH}-r}$ for the CSI feedback. The encoding and decoding functions are jointly designed to reduce the average mismatching between the source CSI Ω and the reconstructed CSI $\hat{\Omega}$. This problem can be represented as

$$(P1): \quad \arg \min_{f_{\text{encode}}/f_{\text{decode}} | \mu, \Psi} d(\hat{\Omega}, \Omega), \quad (7)$$

where $d(\cdot)$ evaluates the mismatching between the source and reconstructed CSI.

III. FRAMEWORK, LEARNING METHODOLOGY, AND PROTOCOL IMPACTS

A. Imperfection-Aware JSCM-based CSI Feedback

In this section, we propose an imperfection-aware learning approach for the training of an autoencoder for enabling JSCM-based CSI feedback. This architecture comprises an encoder deployed at the UE and a corresponding decoder located at the BS. The encoder is responsible for compressing the CSI and generating modulated symbols for transmission, while the decoder reconstructs the CSI from the received symbols. This autoencoder framework effectively performs JSCM, directly mapping the source CSI to complex-valued modulated symbols suitable for transmission over the physical channel, rather than separating source coding, channel coding, and modulation.

The UE first estimates the downlink channel state by using CSI-RS. From this estimated channel state, CSI features are extracted and is fed into the encoder network. The encoder processes these features to produce a latent representation,

usually consisting of a vector of floating-point numbers. These latent variables are subsequently grouped into pairs, where each pair is interpreted as the real and imaginary parts of a complex value. The resulting sequence of complex values is organized into T_{OFDM} subsets, denoted as $\vec{s}_t$ for $t = 0, 1, \dots, T_{OFDM} - 1$. Each subset $\vec{s}_t = \{s_0, s_1, s_2, \dots, s_{S-1}\}$ contains S complex symbols, intended to be mapped onto S allocated subcarriers during the t -th OFDM symbol period. Next, to comply with transmission power constraints, the power of the complex symbols within each subset $\vec{s}_t$ is normalized, on a per-OFDM-symbol basis given by

$$\vec{s}_{\text{Gaussian},t} = \frac{\vec{s}_t}{\sqrt{\mathbb{E}(|\vec{s}_t|^2)}}, \quad (8)$$

where the term $\mathbb{E}\{(|\vec{s}_t|^2)\}$ represents the calculated average power of the symbols within the vector $\vec{s}_t$. This normalization ensures that the transmitted symbols meet power requirements. Finally, the real and imaginary parts of these normalized complex symbols $\vec{s}_{\text{Gaussian},t}$ serve directly as the in-phase and quadrature components for modulation onto the corresponding subcarriers and subsequent transmission over T_{OFDM} OFDM symbols. Here, we omit PUCCH procedure and leaves it to the following.

After propagation through the wireless channel, the BS receives the modulated symbols. By doing channel equalization, the distortion of fading is largely removed. After that, the real and imaginary components of modulated symbols are extracted as floating-point numbers, and fed into the decoder network for CSI reconstruction. A key characteristic of this JSCM approach is the direct processing of modulated symbols by the neural networks. This bypasses the need for explicit quantization and de-quantization steps commonly found in traditional systems employing separate coding and modulation.

It is noteworthy that this autoencoder framework is general and adaptable, permitting implementations using various neural network architectures based on the specific application requirements and CSI characteristics. For our study, we utilize a classical Transformer-based architecture. Specifically, both the encoder and decoder are constructed using Transformer encoder blocks, arranged in a symmetric structure. Each comprises four sequential Transformer block layers. Input CSI samples, initially dimensioned as the number of transmit antennas, the number of receive antennas, real/imaginary components, and the number of subbands, undergo pre-processing. The first three dimensions (transmit antennas, receive antennas, real/imaginary components) are flattened, resulting in an input matrix of shape with flattened features and the number of subbands. In this representation, each subband corresponds to an element in the input sequence. A linear layer embeds this flattened feature dimension into the model's working dimension. Standard Transformer elements, such as positional encoding and multi-head self-attention mechanisms, are retained as per the original Transformer specification.

During the training phase, we insert three distortion emulation functions between the encoder and decoder. These functions, implemented as a series of non-trainable layers, are designed to simulate the effects of residual fading, RF

imperfections, and the Gaussian noise channel. First, the CSI is fed into the encoder, which outputs T_{OFDM} subsets of normalized complex symbols. To facilitate processing, these complex symbols are flattened into a sequence denoted as s_p . This sequence is then passed sequentially through the distortion emulation functions. The initial function emulates residual fading, where it processes the sequence s_p by applying multiplicative distortion and adding noise to each modulated symbol according to the following model:

$$\tilde{s}_{\text{PUCCH,RS}} = (\hat{h}_r s_{\text{PUCCH}}) / (\hat{h}_r + \beta_{CE} n_{RS}). \quad (9)$$

In Eq.(11), $\hat{h}_r$ is a vector representing fading channel, simulated using a sinusoidal function. The constant parameter β_{CE} represents the level of residual fading, influenced by the channel estimation capability. Additionally, $n_{RS} \in C^k$ is a vector representing AWGN experienced by the reference signal, drawn from a complex normal distribution $n_{RS} \sim \mathcal{CN}(0, \sigma^2)$, where σ^2 is the average noise power. This design targets the emulation of channel estimation errors known to be negatively coherent with the SNR.

Subsequently, the sequence undergoes processing to incorporate RF imperfections. This is achieved using a general Rapp power amplifier (PA) model [28], with parameters from 3GPP specification [29]. The resulting signal is described by

$$\hat{s}_{\text{PUCCH,RS}} = \mathcal{F}_{\text{FFT}} \left(\frac{G \mathcal{F}_{\text{IFFT}}(\tilde{s}_{\text{PUCCH,RS}})}{\left(1 + \left| \frac{G \mathcal{F}_{\text{IFFT}}(\tilde{s}_{\text{PUCCH,RS}})}{V_{\text{SAT}}} \right|^{2p} \right)^{\frac{1}{2p}}} \right), \quad (10)$$

where G , V_{SAT} , and p are PA parameters, obtained by a Gaussian distribution with mean respectively equalling to 31.6228, 79.339, 3, and with variance respectively equaling 1, 2, 0.2. In eq.(10), $\mathcal{F}_{\text{FFT}}(\cdot)$ and $\mathcal{F}_{\text{IFFT}}(\cdot)$ denotes the fast Fourier transform (FFT) and the inverse FFT of a vector. Finally, the effects of the Gaussian noise channel are incorporated by adding AWGN to the sequence:

$$\tilde{s}_{\text{PUCCH}} = \hat{s}_{\text{PUCCH,RS}} + n, \quad (11)$$

where the vector $n \in C^k$ represents AWGN noise.

It is important to note that the parameters within the distortion module of residual fading and RF imperfections are highly specific to the operating system. The former values intrinsically depend on the particular pilot and the channel estimation algorithms, while the latter values depend on the receiver's RF implementation. Consequently, establishing a universal parameter set for all scenarios is infeasible. Nevertheless, from a practical deployment perspective, this specificity aligns with the current industry consensus on AI-enabled CSI feedback, where the real-world deployment necessitate an online fine-tuning phase [30]. This phase is crucial for adapting the pre-trained model to local channel distributions and hardware impairments, thereby enhancing performance and facilitating model pruning to reduce computational overhead. Therefore, during the initial deployment phase, the parameters of distortion modules can be set to precisely match the BS's specific channel estimation performance and RF characteristics.

In the subsequent analysis for generating simulation results, we trained the model incorporating only the residual fading and Gaussian noise channel emulation functions. This

configuration was chosen because our link-level simulator (LLS) does not include a model for the PA. However, for our PoC evaluation, which involves a real-world RF chain equipped with a peak shaving technique, we trained the model using all three distortion emulation functions. The training for the PoC specifically incorporated the RF imperfection module configured to account for the effects of a fixed-power peak cancellation scheme, alongside the fading and noise functions, to better reflect the PoC's hardware characteristics. Furthermore, as a potential extension or for different scenarios, other PAPR reduction techniques could be integrated into the model, potentially placed immediately after the RF imperfection module, to create an even more realistic representation of signal distortion. Incorporating the residual fading and RF imperfection modules offers a slight performance enhancement in the low SNR region. More importantly, however, their inclusion serves to align the performance characteristics observed during training with those encountered in the actual implementation environment. This consistency facilitates a more reliable estimation of the CSI reconstruction accuracy or confidence.

Accordingly, the downlink CSI Ω is fed forward sequentially through the encoder, distortion emulation modules, and decoder, before updating the weights in the direction of minimizing the loss. To effectively train the JSCM autoencoder, a dual-loss function is employed, linearly combining the normalized mean square error (NMSE) and the generalized cosine similarity (GCS) via a weighting parameter λ . The loss function is given by

$$\mathcal{L}_{all} = \lambda \mathcal{L}_{NMSE}(\hat{\Omega}, \Omega) + (1 - \lambda) \mathcal{L}_{GCS}(\hat{\Omega}, \Omega), \quad (12)$$

where $\hat{\Omega}$ represents the reconstructed CSI. In Eq.(12), the former part is the NMSE loss $\mathcal{L}_{NMSE}$, represented as

$$\mathcal{L}_{NMSE}(\hat{\Omega}, \Omega) = \frac{\sum_{n=0}^{N_{fCSI}} \sum_{m=0}^{N_{tx}} (\hat{\Omega}_{n,m} - \Omega_{n,m})^2}{\sum_{n=0}^{N_{fCSI}} \sum_{m=0}^{N_{tx}} P_{n,m}^2}. \quad (13)$$

And, the latter part is the GCS loss $\mathcal{L}_{GCS}$, given by

$$\mathcal{L}_{GCS}(\hat{\Omega}, \Omega) = -\frac{1}{N_{fCSI}} \sum_{n=0}^{N_{fCSI}} \left(\frac{\|\hat{\Omega}_n \cdot \Omega_n\|}{\|\hat{\Omega}_n\| \cdot \|\Omega_n\|} \right)^2, \quad (14)$$

where $\|*\|$ is the norm function.

The necessity of this dual-objective approach stems from the distinct characteristics of the two metrics. The NMSE enforces numerical proximity between the reconstructed CSI and the ground truth. The inherent normalization operation is used to eliminate the impact of varying sample energy scales, ensuring stable gradient updates across diverse channel fading conditions. Although NMSE comprehensively captures both magnitude and direction, relying solely on it is empirically insufficient. This is because minimizing the mean numerical error does not strictly guarantee the precise spatial angular alignment of the channel vectors. Conversely, the GCS metric focuses exclusively on the angular relationships between vectors, deliberately ignoring magnitude information. This property is highly effective in maintaining the geometric correctness of the CSI direction. However, optimizing a neural network using only GCS can lead to norm drift, wherein

the magnitude of the reconstructed channel vectors becomes inaccurate. By integrating both metrics, the combined loss function strictly bounds both the absolute numerical error and the spatial directional alignment.

Furthermore, the weighting parameter λ dictates the trade-off between these two objectives. In the evaluations presented in this paper, we establish a universal baseline by setting λ equaling to 0.5, which provides a robust balance suitable for general CSI feedback scenarios. However, from a practical deployment perspective, the selection of λ should be based on optimization tasks to ensure alignment between the training objective and system requirements. For instance, if the primary task is single-user MIMO (SU-MIMO) precoding, the correctness of the spatial direction is of utmost importance, and consequently leading to a preference on increasing the weight for GCS component. On the other hand, for tasks such as multi-user MIMO (MU-MIMO) scheduling, the BS requires accurate channel gains to execute user pairing, thus allocating a higher weight to the NMSE component becomes critical to minimizing magnitude distortion.

To ensure the robustness of the JSCM under fluctuating channel conditions, the noise variance within the distortion emulation layer is not fixed during the training phase. If a model were trained at a static SNR, its performance would inevitably degrade when the operational SNR deviates from the training condition. To overcome this, we employ a variant SNR selection for each training iteration. This method forces the autoencoder to learn latent representations under stochastic noise, thus enabling a single unified JSCM model to maintain performance across diverse SNR regimes. In the following paper, both the hardware PoC and SLS results are based on models trained with SNRs sampled from a uniform distribution spanning 0 to 20 dB.

B. PUCCH Design for JSCM-based CSI feedback

To integrate JSCM into practical cellular networks, the conventional PUCCH protocol requires a structural evolution. In the legacy PUCCH pipeline, the bit serves as the fundamental unit for baseband processing, as depicted on the left side of Fig. 1. It begins with obtaining source bits carrying CSI information using a predefined codebook or an end-to-end CSI compression model proposed in 3GPP release 18 (R18) [11]. A quantizer is deployed to convert the compressed source from floating numbers to bits. This CSI bitstream is subsequently multiplexed with other requisite UCI bits (e.g., CQI, rank indicator, a.k.a., RI) or data bits. The aggregated bitstream subsequently undergoes channel coding, followed by bit-level interleaving to scramble the sequence against channel fading. Then, bit-level rate matching is performed by employing either repetition or puncturing, to align the data payload with the available channel resources. Finally, the resulting bits are modulated and mapped to exact OFDM sub-carriers.

However, in the context of CSI feedback, the raw channel data collected by the UE is inherently a sequence of floating-point numbers representing physical channel features, while the ultimate transmission target to generate analog-like modulated signals for over-the-air transmission. While the legacy

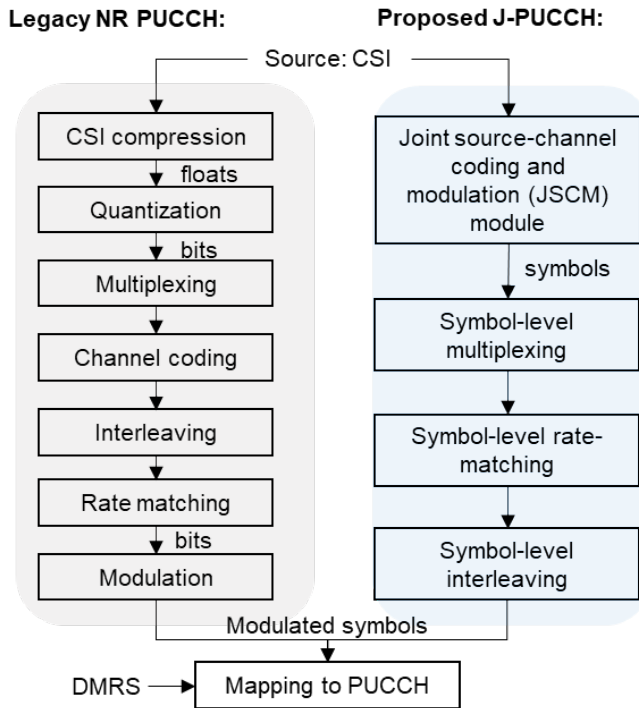


Fig. 1: The architecture of the NR PUCCH and the proposed J-PUCCH procedures.

architecture forces an intermediate step of bit quantization to facilitate standard coding and protocols, the JSCM circumvents this bottleneck by directly mapping the raw CSI into modulated symbols, completely bypassing intermediate bit generation. Consequently, to maintain the indispensable functionalities of the legacy PUCCH within the JSCM framework, a novel procedure is imperative to execute multiplexing, interleaving, and rate-matching directly at the symbol level, termed J-PUCCH.

The proposed J-PUCCH procedure is depicted on the right side of Fig. 1. It is noted that this method has been also proposed in 3GPP technical documentation [31] for more details. First, the JSCM encoder directly outputs a stream of modulated symbols representing the compressed CSI. Concurrently, other data or UCI contents must also be transmitted. Since this information are carried by bits, we can hardly compress them together with the JSCM autoencoder. Therefore, they continue to be processed through the legacy bit-level pipeline to generate their own set of modulated symbols. Subsequently, symbol-level multiplexing is achieved by directly concatenating the JSCM-generated CSI symbols with the other symbols together. To combat channel fading, the aggregated modulated symbols undergoes symbol-level interleaving, where the positions of the complex symbols are scrambled according to a predefined pattern, ensuring that adjacent symbols are dispersed across the time-frequency resources. Following this, symbol-level rate-matching is executed, where the interleaved modulated symbols are punctured or repeated according to their length longer or shorter than the available resources. Finally, similar as legacy PUCCH, the modulated symbols are mapped to exact OFDM sub-carriers. Upon receiving the signal, the BS executes the inverse operations of symbol-

level rate-dematching, de-interleaving, and de-multiplexing, and finally feeding the extracted modulated symbols into the JSCM decoder to recover the CSI. It is worth noting that while this J-PUCCH design presents a promising framework, the exact standardization and physical layer implementation of JSCM remain an active topic of discussion within 3GPP.

However, reusing the existing principles may not fully exploit the potential of JSCM, and more tailored designs of J-PUCCH to optimize JSCM are anticipated. One of key problem is that the autoencoder utilized in JSCM inherently produce a fixed number of output symbols. Conversely, in practical wireless systems, the channel resources allocated to a UE for CSI feedback fluctuate dynamically due to time-varying channel conditions and unpredictable traffic loads. This creates a mismatch between the fixed output size of the JSCM model and the variant resources. Conventionally, this mismatch is resolved via bit-level rate matching, where interleaving inherently causes random puncturing and repetition. While applying this random manner for JSCM forces the autoencoder to learn to recover from information loss at arbitrary positions, which drastically degrades its efficiency. Therefore, a more effective approach is to co-design the J-PUCCH to enable fixed-position puncturing or repetition, thereby alleviating the learning burden on the autoencoder.

Considering that the JSCM can prioritize its output features (to be specified in the next section), we propose a prioritized, symbol-level rate-matching scheme. Specifically, once obtaining the modulated symbols after multiplexing, they are sequentially written into a circular buffer according to their priority. Then, we sequentially map these modulated symbols from the buffer to the available resources. If the allocated resources are constrained, the buffer ensures that rate matching punctures only the least prioritized symbols at the tail end; And conversely, if resources are abundant, the highest-priority symbols at the beginning are sequentially repeated. Crucially, by incorporating this deterministic puncturing and repetition pattern as prior knowledge during the autoencoder training phase, the model is explicitly optimized to handle this specific structure of information loss.

IV. HARDWARE PoC VALIDATION & PERFORMANCE ANALYSIS

A. Hardware PoC Description

We first introduce the details of the hardware PoC test-bed for achieving CSI feedback via PUCCH. The overall architecture and signal processing flow in the test-bed are depicted in Fig. 2. The transmitter emulates the CSI feedback procedure on the UE side that delivers CSI sources, whilst the receiver emulates the CSI reconstruction on the BS side that receives the signal carrying CSI source and recovers it in real-time. On the UE side, RF signals carrying CSI are obtained by sequentially executing AI-based CSI encoding, baseband signal processing, and RF signal processing by personal computer 1 (PC1) running Python, PC2 with LabView, and universal software radio peripheral (USRP) 1, respectively. An inverse procedure occurs on the BS side. The hardware realization is depicted in Fig. 2. The transceivers are realized

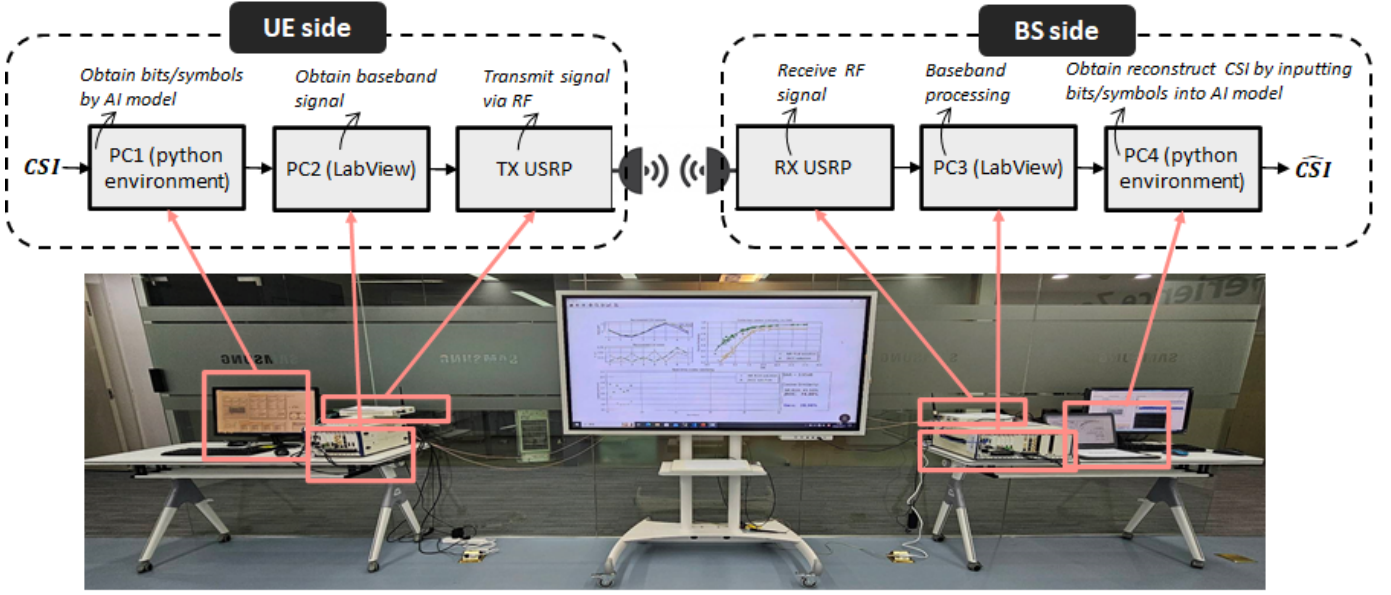


Fig. 2: Architecture, signal processing procedure, and hardware illustration of the PoC.

by USRP-2942, where each USRP is equipped with 1 RF chain supporting a single layer connected by a rubber rod antenna.

We then introduce the realization of both NR-R18 and the proposed JSCM solutions via the developed test-bed. We consider a general NR system implementation with configuration detailed in table I. For NR-R18 solution, the output of PC1 is a sequence of bits that carries compressed CSI obtained via an NR-R18 encoder. After that, PC2 utilizes this bit sequence to generate a baseband signal by sequentially executing the channel coding and modulation mapping. Differently, for the proposed JSCM solution, we obtain the modulated symbols by a JSCM encoder directly on the PC1. It is noted that the AI model and baseband processing of both solutions are executed in a parallel manner. After that, PC2 multiplexes modulated symbols from both solutions together, and maps them to the shared channel resource. The multiplexing guarantees that the transmission of two solutions experiences nearly the same channel condition. Besides payload, we also map a pre-defined preamble sequence at the first symbol for radio frame synchronization and demodulation reference signal (DMRS) at the second and eleventh symbols to support demodulation. Finally, PC2 executes IFFT and deliver the time-domain baseband signal to USRP for the following RF signal generation and transmission.

TABLE I: PoC setting and values.

Setting	Values
Carrier frequency	2.7 GHz
Sub-carrier spacing	30 kHz
Bandwidth	10 MHz
Resource	44 RBs
DMRS type	Type-I
DMRS symbol allocation	[2, 11]

An inverse PUCCH procedure occurs on the BS side. After receiving the signal at PC3, radio frame synchronization is executed by correlation detection between the received signal at the first symbol and the pre-defined preamble sequence. Once synchronized, the least squares (LS) channel estimation, moving averaging de-noising, and linear interpolation

are sequentially executed to estimate the channel impulse response at each resource unit, and then the minimum mean squared error (MMSE) equalizer is used to remove fading at payloads' resource units. Finally, we separate modulated symbols for NR-R18 and JSCM solutions according to the mapping pattern, and execute their decoders independently for CSI recovery.

We utilize the CSI dataset from an open-source library, which is collected in an outdoor environment with 5G commercial network equipment [32]. This data includes CSI with 8 transmitting antenna ports, 4 receiving antenna ports, and 13 sub-bands. To align with the 3GPP's validation methodology [11], we consider precoding vector as our CSI source. We preprocess the full CSI to obtain single-layer precoding vectors by singular value decomposition (SVD) as the feedback target, where the data shape of one sample is $C^{8 \times 13}$. A total of 10,000 data samples are used, with 8,000, 1,000, and 1,000 samples reserved for training, validation, and testing, respectively. It is important to highlight that the realization of the benchmark and the JSCM methods are with the completely same autoencoder architecture - a four-layer Transformer.

We set the hardware PoC in an office that covers approximately 100 square meters and features an open workspace with partitions, desks, and chairs, as well as glass-walled and brick meeting rooms and corridors in the centre area. Testing is conducted during office hours, capturing employees sitting at desks and moving around the open space. We select 10 different locations for testing, including both line-of-sight and non-line-of-sight scenarios with 1 to 2 walls between the BS and UE. During testing, 50 samples per solution are delivered for every slot. The noise level on the receiver side is fixed, and we achieve various SNR levels by adjusting the transmit power on the transmitter side. The SNR is estimated using a conventional NR algorithm deployed on the receiver side, based on the estimated channel response provided by DMRS.

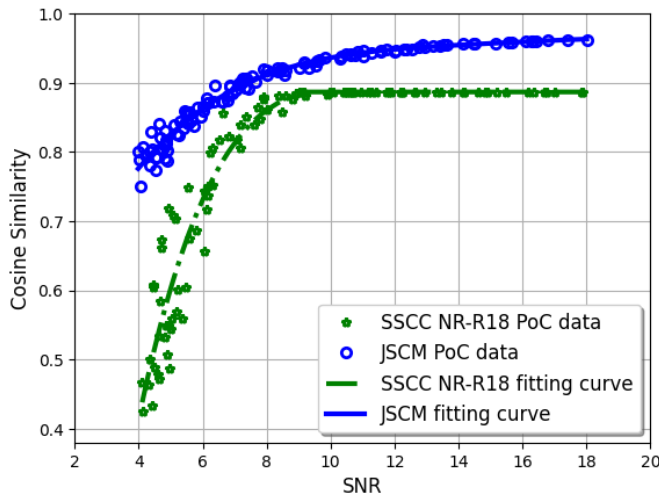


Fig. 3: Statistical GCS accuracy across different SNR regions in PoC field trials.

B. PoC Results and Evaluation

In this sub-section, we present the PoC results to validate performance across different SNR regions. We first plot the GCS accuracy as a function of SNR, spanning a range from 3 to 18 dB in Fig. 3. We plot the NR-R18 solution as the benchmark, which employs an autoencoder model that compresses a CSI sample from 208 floating numbers to 52 bits. We utilize a general CSI feedback configuration at NR setting Polar code with 0.8 coding rate (CR) and quadrature phase shift keying (QPSK) modulation, producing a total of 33 modulated symbols per CSI sample. On the other hand, the JSCM solution utilizes autoencoder models trained under a broad range of noisy channel conditions within the SNR spectrum of [0, 20] dB. It also produces 33 modulated symbols per CSI sample for transmission to guarantee the same overhead as the benchmark.

Given that the practical environment is dynamic, we plot both the PoC results and their fitting curves obtained by a 4th-order polynomial regression function in Fig. 3. As can be seen, the proposed JSCM (blue curve) surpasses the NR-R18 across all SNR regions. The enhancement of the JSCM method is due to its capability to tolerate minor mismatches without significantly impacting CSI reconstruction accuracy, thus it can utilize continuously modulated symbols that carry nearly infinite information. Additionally, the JSCM method exhibits a less pronounced decline in performance as SNR decreases, effectively mitigating the “cliff effect”.

Empirical results demonstrate that even when employing identical neural network architectures, the proposed JSCM framework consistently outperforms the traditional SSCC. Furthermore, it is anticipated that developing specialized models tailored specifically for JSCM will widen this performance gap in the future. The fundamental reasons for JSCM’s superiority are multifaceted, which can be summarized from a principled perspective, that: 1) Rooted in joint coding theory, JSCM allows the model to globally optimize the strategy of source and channel coding; 2) More importantly, by directly integrating modulation into the autoencoder, the model naturally learns to map features into a continuous, Gaussian-like distribution. From an information-theoretic perspective, it is

a well-established principle that a Gaussian distribution of power can achieve the Shannon capacity limit over an AWGN channel, thereby maximizing the theoretical data throughput.

In traditional digital communication systems, adopting Gaussian-distributed continuous modulation is prohibitive. Traditional schemes are strictly constrained by the requirement for lossless, and their limited capacity to handle noise on unquantized continuous symbols would lead to severe error. However, for inherently analog or continuous sources, such as CSI or audio, strict bit-level lossless transmission is unnecessary. For instance, a minor continuous perturbation in CSI values does not significantly degrade beamforming performance, whereas a single bit error in a traditional quantized system could corrupt the entire CSI report. The rapid evolution of AI has introduced unprecedented denoising capabilities. Consequently, in an end-to-end AI transmission system, the distortion introduced by the forced quantization step (i.e., mandatory in traditional digital modulation) often exceeds the distortion caused by the channel noise itself. Therefore, for continuous analog sources, it is imperative to move beyond the rigid shackles of traditional discrete quantization and modulation paradigms, transitioning instead towards AI-driven, unified continuous mapping approaches.

V. CRITICAL ENABLERS FOR JSCM-BASED CSI FEEDBACK

A. Critical Enabler I: Variant-resource adaptation

Reminding the challenge of arbitrary resources allocation for UE CSI feedback, one need a JSCM autoencoder capable of generating variant-length latent features with negligible performance degradation. This refers to a variable-rate communication problem, where several approaches have been explored in the literature, primarily including variable-length quantizer [23] and mask learning [16]. Quantizer-based methods adapt to varying output by dynamically adjusting the quantization resolution, while these methods are exclusively tailored for JSCM given the quantization is omitted. The target requires to accommodate the highly stochastic and fine-grained resource allocations, thus the mask learning direction is more affordable.

With the mask learning direction, we propose an encoder architecture that generates latent sequence with inherent priority levels. This architecture enables a priority-aware rate matching scheme, allowing the JSCM framework to adapt flexibly to arbitrarily configured resource allocations. The core principle is to minimize the loss of critical latent information when puncturing occurs. This is achieved by implicitly structuring the latent representation during training such that the most significant latent features are concentrated within the initial portion of the output sequence, while modulated symbols carrying less critical refinement information are placed later. In contrast to conventional communication systems that typically perform interleaving followed by rate matching primarily for fading mitigation, our method requires a specific processing order to leverage the learned priorities. Therefore, we propose that the UE first performs priority-aware rate matching on the generated modulated symbols, followed by interleaving.

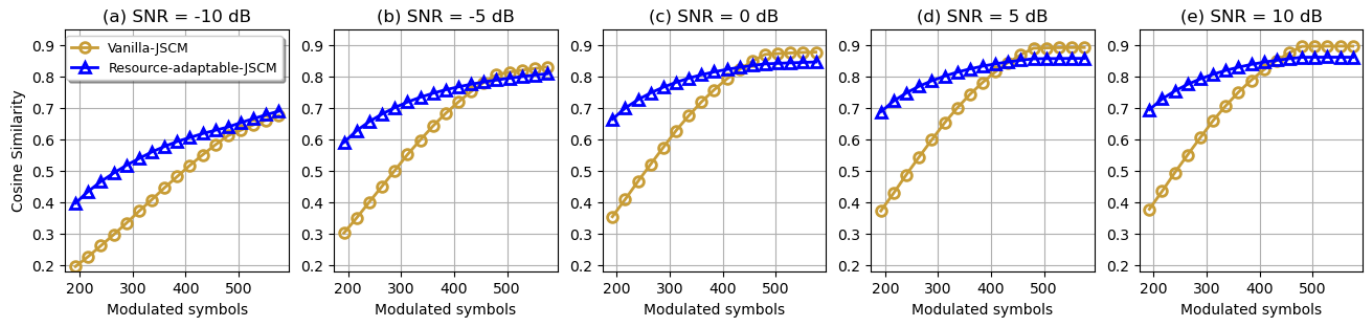


Fig. 4: GCS performance of JSCM versus the different allocated resources with (a-e) the PUCCH SNR of -10, -5, 0, 5, and 10 dB.

Specifically, the rate matching process operates directly on the encoder's output sequence, which intrinsically orders symbols by priority from the highest to the lowest. When puncturing is required (i.e., allocated resources are fewer than the nominal model output), modulated symbols are punctured sequentially starting from the lowest priority up to the required number. Conversely, when repetition is necessary (i.e., allocated resources exceed the nominal model output), modulated symbols are duplicated sequentially starting from the highest priority down to the lowest. This priority-aware rate matching ensures that the final set of transmitted symbols aligns with the available resources while preferentially retaining the most crucial CSI information.

We now detail the training methodology employed to realize an encoder capable of generating prioritized outputs. This method utilizes a data augmentation technique based on random-length puncturing to simulate the variable resource allocation during training. The objective is to implicitly train the model to structure its latent representation such that critical information is concentrated in modulated symbols less likely to be punctured, thereby maximizing information retention even when a portion of the output is discarded. During the training phase, for each input sample, a random puncturing length parameter L_{punct} is generated, where L_{punct} is less than the total number of output modulated symbols produced by the encoder. This parameter signifies the quantity of symbols to be notionally discarded. After the encoder generates the full sequence of output modulated symbols, the last L_{punct} modulated symbols are discarded. Crucially, the subsequent normalization step is performed only over the remaining unpunctured modulated symbols to ensure that transmission power is not allocated those punctured ones. At the decoder, only the unpunctured modulated symbols are received. To maintain the structural integrity required by the decoder's input layer, zero-padding is applied at the positions corresponding to the punctured symbols. By adopting this training strategy, which mimics the targeted rate-matching behavior, the JSCM autoencoder learns to embed priority intrinsically within its output sequence, yielding the desired priority-aware model.

Building upon the preceding PoC validation, which utilized field-collected channel data and adhered to prevailing 3GPP validation standards by targeting the precoding vector for feedback, we successfully demonstrated the foundational performance advantages of the JSCM approach. For the subsequent evaluation of multiple advanced designs, we transition to a

link-level simulation (LLS) platform to model the PUCCH. Furthermore, to facilitate the straightforward computation of downstream downlink throughput in our later analyses, we will consider channels modelled according to the 3GPP Urban Macro (UMa) specification as the feedback target.

A persistent aspiration among MIMO system designers is the acquisition of explicit CSI at the BS, extending beyond merely a single-stream precoding vector. Access to explicit CSI would empower the BS to perform more complex and potentially superior precoding optimizations. However, the feedback of explicit CSI using traditional techniques typically results in prohibitive overhead. The JSCM framework introduced in this work exhibits significant potential for conveying more extensive CSI information at substantially reduced overhead, thereby rendering explicit channel feedback practically attainable. Motivated by this prospect, our subsequent validations will specifically consider the scenario of explicit CSI feedback.

We then present simulation results to demonstrate the efficacy of the proposed resource-adaptable JSCM approach. The CSI data is obtained by UMa channel model with 32 transmitting antenna ports (BS side), 4 receiving antenna ports (UE side), and 13 sub-bands (each covering 4 RBs). In the evaluation, "Vanilla-JSCM" denotes the baseline JSCM model trained without augmentation, where results are obtained by directly applying puncturing or repetition to its standard output. Both models utilize identical network architectures, ensuring equivalent computational complexity. Figure 4 illustrates the GCS as a function of the number of allocated modulated symbols. Performance is plotted under various PUCCH SNRs of -10, -5, 0, 5, and 10 dB. The nominal output size of our encoder is 480 modulated symbols, and the figure displays performance ranging from 192 to 576 modulated symbols.

As expected, the GCS performance generally degrades as the allocated resources decrease below the nominal puncturing regime (480 modulated symbols). When the allocated resources exceed 480, the GCS performance at lower SNRs tends to improve with increasing resources due to the combining gain achieved through repetition. However, performance saturates at higher SNRs, indicating that the upper bound is determined by the inherent information compression capability of the JSCM model rather than channel noise.

Critically, comparing the two approaches across all tested SNRs reveals that the resource-adaptable-JSCM consistently exhibits a more gradual decline in GCS performance as the

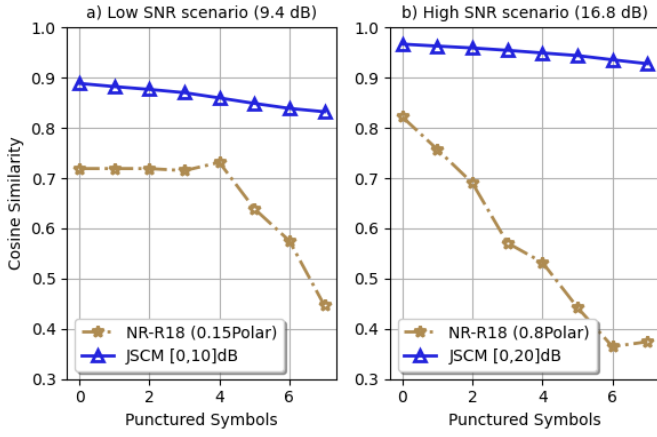


Fig. 5: Statistical GCS accuracy across different numbers of symbols being punctured in PoC field trials, under scenarios of: a) Low SNR; and b) High SNR.

allocated resources decrease compared to the Vanilla-JSCM. This advantage becomes particularly pronounced when significant puncturing occurs. Specifically, when the allocated resources are half of the nominal training resources (240 modulated symbols) or less. In this region, the GCS achieved by resource-adaptable-JSCM is nearly double that of Vanilla-JSCM. This observation strongly suggests that the proposed priority-aware training method successfully enables the JSCM model to learn an effective prioritization of its output symbols, ensuring that essential information is preferentially retained even under severe resource constraints.

Finally, we validate the performance of JSCM's resource adaptation capability through our PoC experiment. Note that this PoC albeit on single-layer precoder vector feedback, with models trained for a nominal output of 33 modulated symbols, with the proposed random-length puncturing data augmentation technique. As illustrated in Fig. 5, the horizontal axis of each sub-figure depicts the number of modulated symbols x being punctured out of the nominal 33 modulated symbols (i.e., $33-x$). It shows that the GCS of the resource-adaptable-JSCM (blue curve) degrades only about 5% even with 8 punctured modulated symbols, while that of the NR-R18 reduces about 30% and 50% in low and high SNR scenarios, respectively. This again highlights the ability of a priority-aware JSCM to effectively capture and preserve the most critical information-bearing modulated symbols during puncturing.

B. Critical Enabler II: Low PAPR Waveform

In the current implementation of our JSCM framework, the autoencoder generates latent representations as floating-point numbers. These latent variables are paired and subsequently mapped to complex-valued modulated symbols intended for transmission. Critically, power normalization is applied collectively across the entire bandwidth of complex symbols constituting one OFDM symbol. This strategy is employed to maximize the information carried by the wave, resulting in a constellation diagram whose powers approximate a Gaussian distribution as shown in Fig. 6(a). This approach offers two principal advantages: 1) From an information-theoretic perspective, a Gaussian input distribution is known to approach

the Shannon capacity limit for an AWGN channel, thereby maximizing theoretical data throughput; 2) This normalization permits the model to implicitly allocate power across symbols based on the importance of the underlying latent features, effectively embedding an intelligent power control.

However, uplink transmissions in cellular systems predominantly utilize discrete Fourier transform-spread OFDM (DFT-s-OFDM) waveforms specifically to mitigate the high PAPR inherent in conventional OFDM. DFT-s-OFDM introduces a DFT precoding stage prior to the inverse fast Fourier transform (IFFT). This precoding effectively spreads the energy of each modulation symbol across the entire bandwidth, rather than confining it to a single subcarrier, which consequently lowers the PAPR of the time-domain signal. Nevertheless, the PAPR of the resulting time-domain DFT-s-OFDM signal remains directly correlated with the PAPR of the frequency-domain symbols input to the DFT precoder. Specifically, the ratio between the maximum and minimum power among these input symbols significantly influences the final PAPR. Consequently, the Gaussian-like constellation produced by our current per-OFDM-symbol normalization scheme, characterized by its substantial amplitude variations, inherently leads to a high PAPR when transmitted using DFT-s-OFDM. An extreme countermeasure involves normalizing the power on a per-symbol basis, yielding a constant-amplitude, phase-only modulation scheme as shown in Fig. 6(f). In details, each modulated symbol s_k within a subset $\vec{s}_t = \{s_0, s_1, s_2, \dots, s_{S-1}\}$ is normalized by

$$s_{\text{circle},k} = \frac{s_k}{\sqrt{|s_k|^2}}. \quad (15)$$

While this achieves the lowest possible PAPR, comparable to that of conventional QPSK with DFT-s-OFDM, it incurs a substantial penalty in information-carrying capacity, potentially reducing it by nearly half.

To mitigate the high PAPR challenge arising from the Gaussian-like constellation while retaining the benefits of JSCM, we propose a modulation shaping strategy aimed at reconstructing the power variance of the transmitted symbols. This yields what we term annular-Gaussian modulation, characterized by a constellation where symbol power is concentrated within an annular region around the unit circle. The process commences with the initial per-OFDM-symbol power normalization, applied in the baseline Gaussian approach, ensuring the average power of all modulated symbols within each OFDM symbol is unity. Subsequently, while preserving the total power allocated to the OFDM symbol, the power variance among the constituent modulation symbols within the OFDM symbol is deliberately reduced.

Specifically, given the set of complex symbols $\vec{s}_{\text{Gaussian},t}$ for OFDM symbol t resulting from the per-OFDM-symbol normalization obtained by Eq.(8), the corresponding annular-Gaussian modulated symbols $\vec{s}_{\text{annular},t}$ can be obtained by

$$\vec{s}_{\text{annular},t} = \vec{s}_{\text{Gaussian},t} \odot \sqrt{\frac{1 + \alpha_{\text{scale}}(|\vec{s}_{\text{Gaussian},t}|^2 - \mathbb{E}(|\vec{s}_{\text{Gaussian},t}|^2))}{|\vec{s}_{\text{Gaussian},t}|^2}}. \quad (16)$$

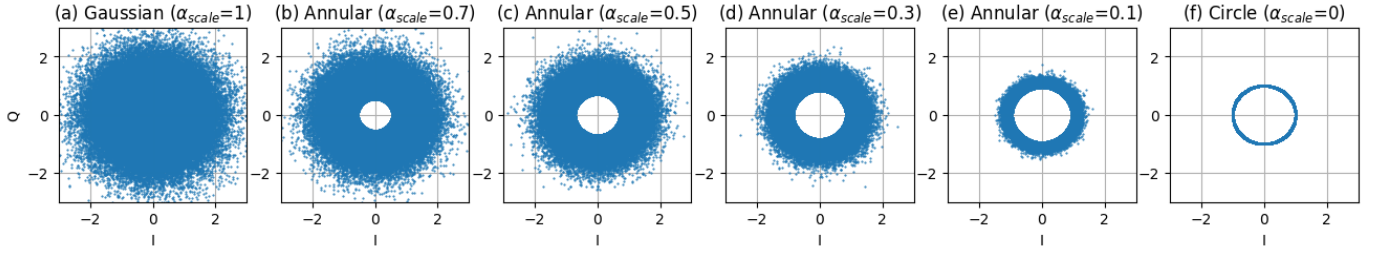


Fig. 6: Modulation constellation of (a) Gaussian modulation; (f) Circle modulation; and (b-e) Annular-Gaussian modulation with the power-variance scaling factor α_{scale} equaling to 0.7, 0.5, 0.3, and 0.1.

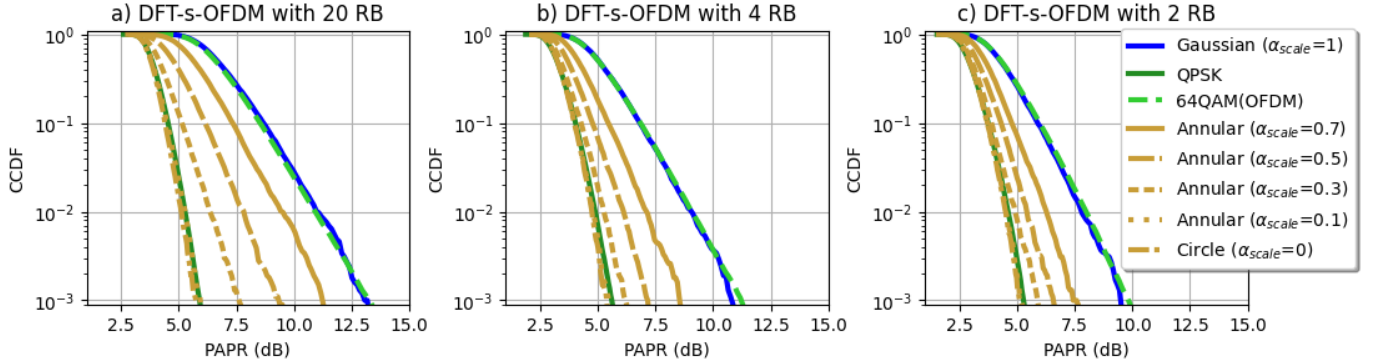


Fig. 7: Comparison of PAPR performance (CCDF) for Gaussian modulation, circle modulation (i.e., constant modulus), and the proposed annular-Gaussian modulation using various power-variance scaling factor α_{scale} equaling to 0.7, 0.5, 0.3, and 0.1, with DFT-s-OFDM under a) 20 RB, b) 4 RB, and c) 2 RB.

Here, $\odot$ represents element-wise multiplication. The expectation $\mathbb{E}(|\vec{s}_{\text{Gaussian},t}|^2)$ is equal to 1 due to the preceding normalization step in Gaussian modulation. In the transformation described by Eq.(16), the core idea is to scale the deviation of each modulated symbol's power from the established mean power. First, this power deviation, calculated as $|\vec{s}_{\text{Gaussian},t}|^2 - \mathbb{E}(|\vec{s}_{\text{Gaussian},t}|^2)$ for each modulated symbol in the vector, is obtained. This deviation term represents the variance component of the power for each symbol relative to the mean. Hence, it can be regarded as a set of modulated symbols with zero mean power. Then, this power deviation is scaled by a power-variance scaling factor $\alpha_{scale} \in [0, 1]$. A smaller α_{scale} results in a greater reduction of the power variance. Following the scaling of the deviation, the unit mean power (i.e., equaling to 1) is added back, yielding the target power magnitude for each symbol in the annular-Gaussian constellation. Finally, to achieve this target power while preserving the phase of the original symbol, the original complex symbol $\vec{s}_{\text{Gaussian},t}$ via Gaussian modulation is multiplied by a real-valued scaling factor, derived from the root squared ratio of its target power to its original power. This element-wise multiplication ensures that the resulting annular-Gaussian symbols in $\vec{s}_{\text{annular},t}$ maintain the phase information of $\vec{s}_{\text{Gaussian},t}$ but possess modified magnitudes such that their squared magnitudes equal the calculated target powers.

The principal benefit of this annular-Gaussian approach lies in its inherent flexibility to control the constellation's power distribution via the parameter α_{scale} . Constellations generated using the annular-Gaussian approach with varying power-variance scaling factors α_{scale} of 0.7, 0.5, 0.3, and 0.1 are depicted in Fig. 6 (b), (c), (d), and (e), respectively. It is observed that the variance of the power associated with the modulated symbols diminishes as the power-variance scaling

factor α_{scale} , spanning a range between that of the Gaussian and circle modulation. In practical deployments, the parameter α_{scale} can be adjusted to achieve a flexible trade-off between PAPR reduction and CSI reconstruction performance, catering to specific hardware constraints and PA thresholds of the UE. To effectively balance this trade-off, it is essential to optimize the waveform by maximizing end-to-end performance within the constraints of specific hardware limitations.

The complementary cumulative distribution functions (CCDFs) of the PAPR for the Gaussian modulation, circle modulation (i.e., constant modulus), and annular-Gaussian modulation (with α_{scale} equaling 0.7, 0.5, 0.3, and 0.1) are presented in Fig. 7. Additionally, we also include the CCDF of PAPR for QPSK and 64QAM for comparison. For comparative purposes, the CCDFs of PAPR for QPSK and 64QAM are also included. Within each sub-figure, all PAPR results pertain to the DFT-s-OFDM waveform, with the exception of the 64QAM using an OFDM waveform, which serves to illustrate a relative upper tolerance for PAPR.

It is crucial to recognize that PAPR performance is intrinsically linked to the allocated bandwidth. In typical operational scenarios, BS tends to assign narrower bandwidths to cell-edge UEs. This practice stems from the necessity for cell-edge UEs to operate their PAs close to saturation to maximize their link budget. A smaller allocated bandwidth results in a lower PAPR, which in turn allows for a higher average transmitted power, thereby enhancing communication reliability for these UEs. To demonstrate, we plot PAPR curves in scenarios of allocating 20, 4, and 2 resource blocks (RBs) in Fig. 7. We can observe that the PAPR of the waveform decreases naturally alongside the reduction in allocated RBs. As shown in Fig. 7 (c), with α_{scale} equaling 0.5, the annular-Gaussian modulation contributes to a PAPR with about 6.5 dB, that is only 1.5

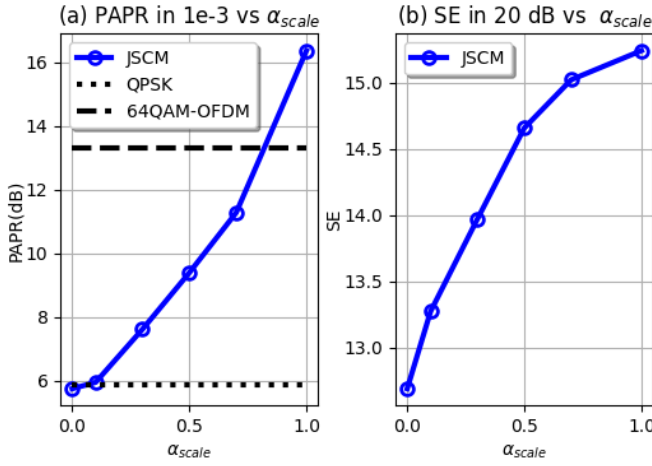


Fig. 8: Comparison of (a) PAPR, and (b) spectral efficiency performance versus power-variance scaling factor α_{scale} .

dB higher than the minimal QPSK setting. This makes JSCM implementable in practical cellular systems. Consequently, in narrow-band scenarios, the system can afford to employ a modulation shape with a larger α_{scale} , which allows the UE to maximize the CSI reconstruction performance while still comfortably satisfying the stringent PAPR constraints of the RF front-end.

For annular-Gaussian modulation, when $\alpha_{scale} \approx 1$, the power distribution of modulated symbols closely resembles the original Gaussian profile. In contrast, when $\alpha_{scale} \approx 0$, the individual modulated symbol powers converge towards unity (i.e., $|s_k|^2 \approx 1, s_k \in \vec{s}_{annular,t}$), causing the constellation to approximate a constant modulus signal, thereby yielding a significant reduction in PAPR. This mechanism provides a configurable trade-off between feedback accuracy and PAPR, allowing adaptation to specific system requirements, such as configuring a larger bandwidth requiring a smaller PAPR.

Figure 8 presents a comparison of PAPR and spectral efficiency performance as a function of the power-variance scaling factor α_{scale} . The spectral efficiency is computed assuming 4 transmission layers employing a ZF precoder, under conditions of 20 dB SNR for both the PUCCH (carrying CSI feedback) and the PDSCH (for downlink data transmission). The results indicate that increasing α_{scale} enhances spectral efficiency due to more accurate CSI reconstruction, albeit at the expense of a higher PAPR. Conversely, reducing α_{scale} lowers the PAPR, but incurs some degradation in spectral efficiency performance. Notably, the PAPR exhibits an exponential increase with α_{scale} , whereas the spectral efficiency demonstrates a logarithmic increase. Consequently, setting α_{scale} to 0.5 reduces the PAPR of Gaussian-JSCM by 56% (at a CCDF of $1e-3$), while incurring only a 4% loss in spectral efficiency performance.

Subsequently, focusing on the case where α_{scale} equals 0.5, Figure 9 evaluates the GCS and SE performance across varying PUCCH SNR levels. The spectral efficiency performance achieved with the e-Type2 codebook is included as a reference benchmark. The results demonstrate that, despite the 56% PAPR reduction relative to Gaussian-JSCM, the annular-Gaussian modulation with $\alpha_{scale} = 0.5$ exhibits only

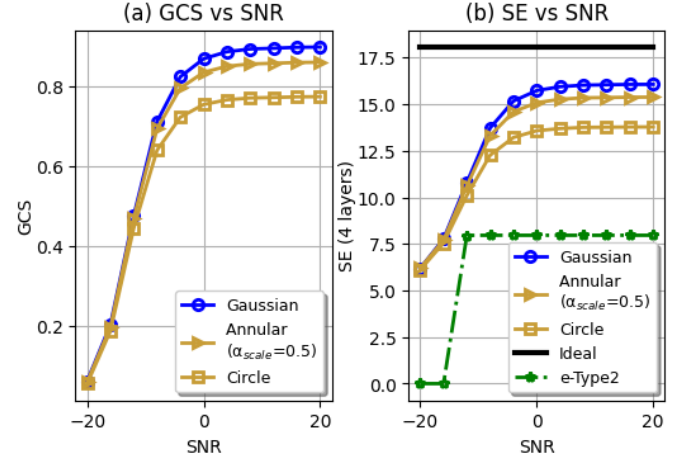


Fig. 9: Comparison of (a) GCS and (b) SE performance versus PUCCH SNR.

a marginal decrease in both GCS and spectral efficiency performance compared to the Gaussian-JSCM approach. Importantly, this configuration still achieves a substantial spectral efficiency gain exceeding 80% compared to the e-Type2 codebook solution across the evaluated range of PUCCH SNR.

C. Critical Enabler III: CQI estimation

In codebook-based CSI feedback, the UE plays a crucial role in facilitating downlink adaptation by estimating the channel quality and providing feedback. Initially, the UE measures reference signals to gather essential CSI. Specifically, interference levels are typically estimated using zero-power CSI-RS (ZP-CSI-RS), while the downlink channel matrix is estimated via non-zero-power CSI-RS (NZP-CSI-RS). Subsequently, the UE performs a codebook search procedure. This involves evaluating the spectral efficiency achievable by applying each candidate precoding vector from a predefined codebook to the estimated downlink channel matrix. The UE identifies the precoding vector that maximizes this spectral efficiency. Combining the channel gain associated with the selected precoder and the previously estimated interference, the UE calculates the resulting Signal-to-Interference-plus-Noise Ratio (SINR). This SINR value is then mapped to a corresponding CQI, which is reported back to the BS. At the BS, a link adaptation (LA) process utilizes the reported CQI, along with other factors such as traffic load and historical data, to select an appropriate modulation and coding scheme (MCS) for subsequent downlink transmissions. Within the scope of this paper, our primary focus is on the CSI feedback mechanism itself. Consequently, the subsequent analysis will proceed based solely on the MCS directly associated with the UE-reported CQI, without delving into the LA process.

However, a significant challenge arises within the proposed JSCM architecture: the UE cannot readily compute the effective downlink channel gain in the same manner as in conventional codebook-based systems. This difficulty is caused by two primary factors inherent to the JSCM approach. Firstly, unlike codebook schemes where the UE stores the complete set of predefined precoding vectors, UE lacks knowledge of the potential reconstruction inaccuracies or distortion introduced by the decoder operating at the BS. Secondly, as the latent

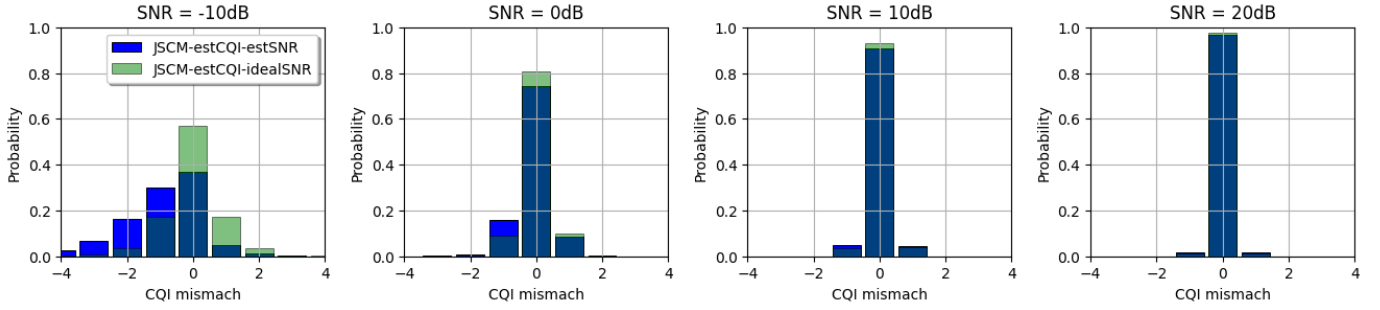


Fig. 10: Statistical distribution of CQI estimation mismatches under (a-d) PUCCH SNR of -10, 0, 10, and 20 dB.

representation traverses the uplink channel during the feedback process, it is inevitably corrupted by channel noise. The UE, operating locally, cannot directly ascertain the precise extent of this noise-induced distortion.

To overcome this impediment, we introduce a CQI estimation methodology predicated on UE. The successful implementation of this technique requires the BS to furnish the UE with two auxiliary information: 1) The specific decoder configuration or model parameters utilized by the BS, which can be conveyed to the UE during the radio resource control (RRC) setup phase; 2) The real-time uplink SNR, which the BS estimates based on uplink demodulation reference signals (DMRS) and subsequently signals to the UE via downlink control information (DCI). Operationally, during the CSI feedback phase, once the UE computes the latent representation of the CSI using its local encoder, it utilizes the received uplink SNR value to artificially synthesize noise, and then intentionally added to the clean latent representation. This procedure thereby emulate the perturbations the representation is expected to undergo during transmission over the actual uplink channel. Subsequently, this deliberately noise-corrupted latent representation is fed into the locally available decoder. The output of this local decoding process yields an estimate of the CSI as it would likely be reconstructed at the BS. With this estimated reconstructed CSI, the UE can then proceed to calculate the CQI using conventional methods, analogous to the final steps performed in codebook-based systems.

We consider a prevalent uplink SNR estimation algorithm commonly employed in existing 4G/5G systems. At the BS, the noise power $P_{r,t}^N$ is estimated utilizing DMRS performed individually for each antenna port r and OFDM symbol t . In detail, $P_{r,t}^N$ is estimated using a second-order difference method applied to the uplink channel matrix, calculated as

$$P_{r,t}^N = \frac{1}{6|\mathcal{K}|} \sum_{k \in \mathcal{K}} |2H_{r,t}^{\text{UL}}(k+1) - H_{r,t}^{\text{UL}}(k) - H_{r,t}^{\text{UL}}(k+2)|^2, \quad (17)$$

where $H_{r,t}^{\text{UL}}(k)$ denotes the uplink channel response estimated at subcarrier index k for antenna r and OFDM symbol t . $\mathcal{K}$ represents the set of subcarrier indices k , and $|\mathcal{K}|$ denotes its cardinality. Subsequently, the overall SNR is computed by averaging the per-antenna, per-symbol SNR values across the antenna ports set $\mathcal{R}$ and OFDM symbols set $\mathcal{T}$. Assuming normalized signal power, this uplink SNR is given by

$$\text{SNR}_{\text{UL}} = \frac{1}{|\mathcal{T}||\mathcal{R}|} \sum_{t \in \mathcal{T}} \sum_{r \in \mathcal{R}} \frac{1}{P_{r,t}^N}. \quad (18)$$

Note that estimating noise power by the second-order difference method in Eq. (17) assumes that the channel responses across three adjacent DMRS-carrying subcarriers are identical. However, this assumption does not perfectly hold due to the frequency-selective fading. The channel variations across adjacent subcarriers are captured by the difference operation and accumulated into the noise power statistics. Consequently, the estimated noise power could be slightly higher than the true one, directly resulting in an underestimated SNR.

As observed in the performance curves, the “JSCM-estCQI-estSNR” scheme tends to conservatively underestimate the CQI, particularly in the low SNR regime. This phenomenon can be attributed to the inherent mechanism of the applied practical SNR estimation algorithm. Specifically, the noise power is calculated using the second-order difference method across subcarriers bearing Demodulation Reference Signals (DMRS). This technique inherently assumes that the channel frequency response remains completely flat across three adjacent DMRS subcarriers. In realistic frequency-selective fading environments, however, this assumption is easily violated. The natural channel fluctuations between adjacent subcarriers are inevitably captured by the difference operation and erroneously aggregated into the noise power statistics. This leads to an overestimation of the noise variance and, consequently, an underestimation of the effective SNR, which forces the system to report a lower, more conservative CQI value.

A critical deployment challenge for this method is the assumption that the UE possesses knowledge of the BS decoder parameters. In practical cellular networks with multiple vendors, a BS vendor’s specific AI model is strictly protected intellectual property (IP), making direct parameter sharing infeasible. Fortunately, recent consensus within 3GPP on AI-enabled CSI feedback has paved the way for a standardized resolution, where the prevailing approach involves defining a standardized cross-vendor reference decoder [30, 33]. UEs from any manufacturer can utilize this openly specified reference decoder to evaluate the reconstruction quality and compute the CQI, thereby ensuring seamless cross-vendor interoperability without compromising the proprietary, optimized implementations residing at the BS. This establishes the practical applicability of computing CQI via direct CSI reconstruction at the UE [30].

Alternatively, to mitigate the computational overhead of running the full reference decoder at the UE, one may train a

proxy model at the UE-side to estimate reconstructed CSI [9], or even CQI directly [30]. To adapt this proxy-based paradigm to the proposed JSCM framework, the operating SNR must be introduced as an additional feature input to the proxy model, ensuring it captures the channel-induced distortions inherent to joint coding. The direct estimation proposed in this work may provide a high-fidelity assessment of the CQI due to the exactly mirror of the CSI reconstruction, while the proxy-based approaches offer lower computational complexity but inevitably introduces auxiliary estimation errors due to the approximation of the proxy network. While, all methods represent viable and standard-compliant candidates for future 6G networks [30], presenting a clear engineering trade-off. Due to the limited space, we consider the proxy-based extension as future work.

We subsequently evaluate the performance of our proposed solution. The evaluation is deployed with a dual link-level simulation framework to assess the CQI determined via the previously outlined procedures. This framework involves: (i) simulating the uplink to model SNR estimation at the BS; (ii) calculating downlink SINR by the noise-corrupted latent representation (i.e., corrupted with noise following BS-estimated SNR) at the UE, and mapping to a CQI value according to a predefined table; and (iii) modeling the transmission of this CSI over the PUCCH from the UE to the BS via JSCM, and calculate the exact downlink SINR via reconstructed CSI. The performance is benchmarked against three CQI derivation scenarios. The first scenario, designated “JSCM-estCQI-estSNR”, represents estimated CQI derived from adding artificial noise following the estimated uplink SNR at BS, reflecting practical estimation inaccuracies. The second, termed “JSCM-estCQI-idealSNR”, refers to the estimated CQI obtained by adding artificial noise following the exact uplink SNR, thereby isolating the impact of SNR estimation errors. The third, named as “JSCM-idealCQI”, involves a calculation performed at the BS, which utilizes a genie-aided downlink channel matrix with the actual CSI received after JSCM, serving as a reference point for evaluating the precoder performance achievable with the ideal CQI. It is noted that, for simulation simplicity, the DCI transmission is omitted.

We next proceed to evaluate the accuracy of the CQI estimation. The evaluation metric employed is the CQI mismatch, calculated as the difference between the estimated CQI and the actual CQI (“JSCM-idealCQI”). A negative mismatch value signifies that the estimated CQI is lower than the actual CQI, while a positive value indicates the opposite. Figure 10 illustrates the distribution of CQI mismatch for estimated CQI under various uplink SNR conditions. We present the distributions for two scenarios of “JSCM-estCQI-estSNR” and “JSCM-estCQI-idealSNR”. It is observable that the estimation error magnitude increases as the SNR decreases. Notably, for the SNR > 10 dB regime, the rate of perfectly accurate CQI estimation exceeds 90%, and occurring mismatch instances typically correspond to an error of only one CQI level. Interestingly, in the low SNR region, the “JSCM-estCQI-estSNR” scheme exhibits a propensity to underestimate the CQI relative to the ground truth value. This phenomenon can be attributed to the mechanism of the applied SNR estimation

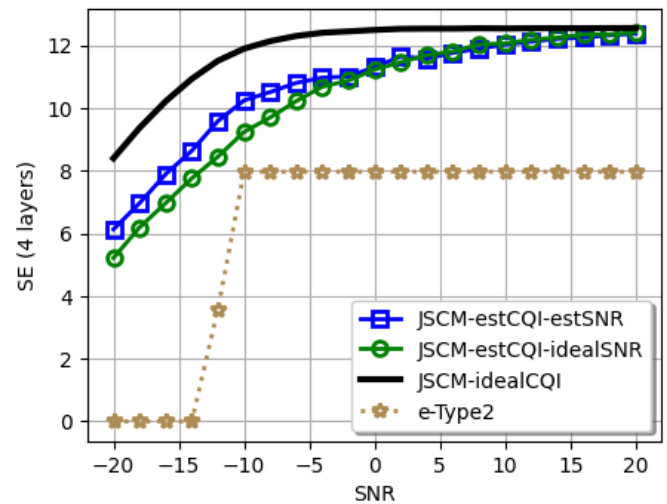


Fig. 11: Compare spectral efficiency under a 4-layer transmission of JSCM with the genie-aided CQI selection, the estimated CQI based on genie-aided PUCCH SNR, and the estimated CQI based on the estimated SNR.

algorithm, where fading is captured and accumulated into the noise power statistics by the second-order difference method. This increased noise power leads to an underestimation of the SINR and consequently the CQI.

Finally, we evaluate the effect of CQI inaccuracies on downlink spectral efficiency. We operate under the assumption that the BS strictly adapts its MCS according to the reported CQI. Under this, we consider CQI overestimation (i.e., estimated CQI > actual CQI) is assumed to result in complete transmission failure, yielding zero spectral efficiency. In contrast, CQI underestimation results in spectral efficiency degradation being calculated according to the rate supported by the estimated CQI.

Figure 11 demonstrates that the degradation in spectral efficiency compared to the ideal CQI selection case (“JSCM-idealCQI”) becomes more pronounced as the SNR decreases. This loss primarily stems from the transmission failures caused by instances of CQI overestimation. We also observe that the upper bound of spectral efficiency of JSCM is slightly reduced compared to that shown in Fig. 9, decreasing from approximately from 14 to 12.5 bits/Hz. This reduction is attributed to the inherent limitation imposed by the CQI table, where samples whose true achievable spectral efficiency exceeds the maximum supported by the largest MCS are capped at the rate of this largest MCS. Conversely, this phenomenon rarely occurs with the codebook-based solution, resulting in a spectral efficiency performance similar to that depicted in Fig. 9. Notably, at low SNR values, “JSCM-estCQI-estSNR” achieves even better spectral efficiency than “JSCM-estCQI-SNR,” despite its CQI estimates being less precise. This outcome is attributed to the tendency of “JSCM-estCQI-estSNR” to conservatively underestimate CQI in the low SNR regime (as shown in Fig. 10), thereby mitigating the penalty of CQI inaccuracy associated with MCS degradation.

Our findings herein reveal that inaccurate CQI estimation is a primary factor contributing to the performance degradation of the JSCM. Despite this, improving UE-based CQI estimation seems to offer diminishing returns, as even more

precise uplink SNR forecasting fails to significantly boost performance. This limitation arises because SNR estimation inherently relies on statistical distributions. Even a statistically perfect estimation cannot capture the instantaneous channel and noise conditions encountered during an actual JSCM transmission, leading to unavoidable errors. A potential avenue for future work to overcome this limitation involves shifting CQI estimation to the BS, allowing it to assess the instantaneous channel pertinent to the JSCM transmission. This could potentially mitigate the mismatch caused by statistical estimation. However, this approach introduces its own challenges, such as acquiring accurate interference levels perceived by the UE, necessitating further in-depth investigation.

D. Compatibility with Digital Baseband

While the proposed JSCM architecture directly maps CSI features into modulated symbols with a continuous-like distribution, it is fundamentally executed within a digital processing unit. The primary challenge for deploying this architecture in current digital cellular systems lies in hardware precision mismatch. Standard neural network accelerators output data in 32-bit floating-point (Float32) format, whereas conventional digital baseband interfaces typically operate with constrained finite-precision formats, such as 8-bit floating-point (Float8). To integrate JSCM seamlessly into standard digital chips, a lightweight and practical solution is to quantize the autoencoder with the same hardware precision of digital baseband hardware. This can be achieved by incorporating quantization constraints directly into the autoencoder's training phase. This method natively aligns the AI output with hardware specifications, averting compatibility issues at the source.

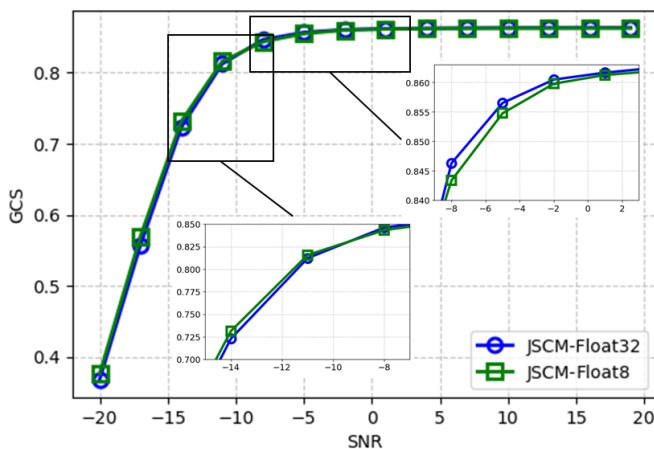


Fig. 12: Comparison of cosine similarity of JSCM with Float32 and Float8 formats.

To evaluate the viability of this integration, we simulated the approach under Float32 and Float8 formats. As shown in Fig. 12, experimental results reveal that the overall performance gap induced by quantization is marginal, with the cosine similarity of the recovered CSI varying by approximately 0.1% between the Float32 and Float8 formats. In the high SNR region, the Float32 symbols outperform the 8-bit counterparts, given that the quantization inevitably discards feature information. Conversely, in the low SNR region, the Float8

scheme exhibits better performance, because quantizing the continuous symbols into discrete levels introduces a degree of inherent noise resilience, similar to how QAM resist noise perturbations.

VI. CONCLUSION

This paper addressed the critical challenge of efficient CSI acquisition in contemporary wireless systems, focusing on translating the concept promise of JSCM into a practical and deployable solution. We presented a comprehensive framework for JSCM-based CSI feedback, which included a learning methodology designed to be robust against real-world channel impairments. The efficacy of this framework was substantiated through the deployment of a hardware PoC platform, which experimentally validated the proposed method's superior performance in realistic operational environments. Furthermore, foundational protocol designs were introduced to facilitate the seamless integration of this advanced CSI feedback mechanism into practical cellular network architectures. A core contribution of this work was the resolution of three pivotal system-level challenges that have previously hindered the widespread adoption of JSCM. Firstly, to manage variant resource allocation, a priority-driven architecture was devised enabling a single model to flexibly adapt to arbitrary resource assignments. This approach demonstrated remarkable adaptability, yet achieving an approximate 90% performance enhancement when resources were halved compared to training. Secondly, the challenge of high PAPR inherent in JSCM-generated waveforms, was mitigated through an innovative modulation shaping technique. This method strategically reallocated information content between signal phase and amplitude, offering a reconfigurable PAPR profile. An annular-Gaussian configuration achieved comparable PAPR with a mere 1% CSI accuracy degradation, thereby providing a flexible trade-off for diverse power amplifier constraints. Thirdly, the difficulty in accurate CQI estimation at the UE was overcome. A novel methodology was introduced where the UE emulates the CSI reconstruction process anticipated at the BS by synthesizing potential uplink channel distortions. This allowed for a reliable estimation of the effective downlink channel gain, maintaining performance parity at high SNRs and exhibiting a performance loss of up to 16% as SNR decreased. Collectively, the synergistic integration of these technical and protocol-level solutions has culminated in a JSCM-based CSI feedback system poised for future 6G networks. Even when accounting for the practical considerations addressed by these three enabling designs, the system demonstrated a significant potential to achieve throughput improvements exceeding 50% compared to conventional codebook-based approaches in high SNR region.

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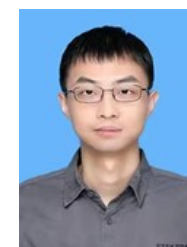
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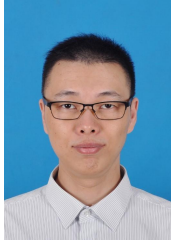
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